Turbulence Model Effects on VOF Analysis of Breakwater Overtopping during the 2011 Great East Japan Tsunami

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ABSTRACT: Numerical simulations are used to investigate the mechanisms of failure of the Kamaishi breakwater during the 2011 tsunami. Delft 3D is used to simulate the propagation of the tsunami as a shallow water wave into Kamaishi Bay, in order to estimate the time progression of water levels on each side of the breakwater. OpenFOAM's InterFOAM VOF solver with RANS k-ε turbulence is then used to simulate the unsteady overtopping of a solid breakwater caisson atop a porous rubble mound, and to calculate the total forces exerted on the breakwater by the flow. Interestingly, the simulated overtopping flow is sensitive to the turbulence model used. In comparison with experiments, the standard OpenFOAM k-ε turbulence model predicts too much entrainment of air below the overtopping jet into the fluid, thus pulling the jet quite close to the landward wall of the breakwater. For better agreement with experimental results, reduction of the turbulent eddy viscosity near the air-water interface proves necessary. Results of the simulations indicate that failure of the breakwater due to sliding or toppling was unlikely. However, the bearing stress at the heel of the caisson increased above its allowable limit due to both the water level difference across the breakwater, and due to the overtopping jet pulling away from the landward side of the caisson as the surcharge height above the caisson increased. In addition to the scour-induced failure of the rubble-mound foundation reported by other researchers, the present research indicates that measures to prevent geotechnical punching failure are also necessary to ensure integrity of the structure during overtopping events.

KEY WORDS: Tsunami, Breakwater, Overtopping, Bearing stress, Turbulence.

1 INTRODUCTION

The Great East Japan Tsunami of March 11, 2011 caused widespread devastation, though in some locations its impact was mitigated by coastal defensive structures such as the bay-mouth breakwater in Kamaishi (Takahashi et al, 2011), the largest and deepest breakwater in the world, constructed exclusively for mitigating tsunami impacts. Figure 1 shows the location of the Kamaishi breakwater, and Figure 2 shows a profile-view geometry of one of this breakwater's typical deep-water caissons and its rubble mound foundation. During the tsunami, the breakwater was overtopped. After the event, many caissons were no longer visible above the waterline, and sonar surveys later showed these to be lying on the seabed, having fallen off their rubble mound foundations in the landward direction (Arikawa et al, 2012). Other caissons were still emergent from the water, but were no longer standing vertical.

In addition to showing caisson displacement, sonar measurements showed significant erosion of the rubble-mound foundation. Arikawa et al (2012) carried out laboratory experiments to replicate the overtopping of the Kamaishi breakwater, and concluded that hydrodynamic scour was the fundamental cause of caisson displacement. Scour due to the overtopping jet plunging down the landward side of the caisson dug a trough beside the heel of the caisson, eventually causing the caisson to tumble.

Additionally, the 1-m gap between caissons may have caused a flow strong enough to cause scour between caissons as well.

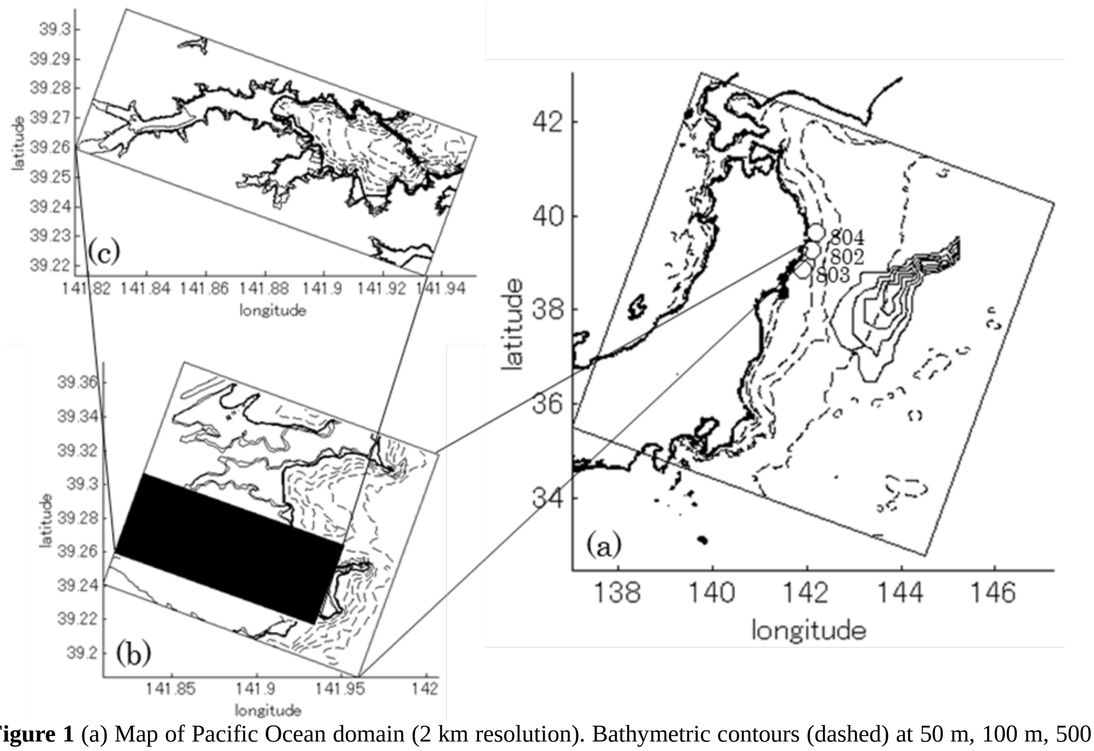

Figure 1 (a) Map of Pacific Ocean domain (2 km resolution). Bathymetric contours (dashed) at 50 m, 100 m, 500 m, 1000 m, and 5000 m. Tsunami initial height contours (solid) at 2 m, 4 m, 6 m, and 8 m. Open circles are PARI GPS buoys 802, 803, and 804. (b) Map of Kamaishi area coast (200 m resolution). Bathymetric contours (dashed) and elevation contours (solid) at 20 m intervals (topography above 40 m omitted for clarity). (c) Map of Kamaishi Bay domain (20 m resolution). Bathymetric contours (dashed) and elevation contours (solid) at 10 m intervals (topography above 40 m omitted for clarity).

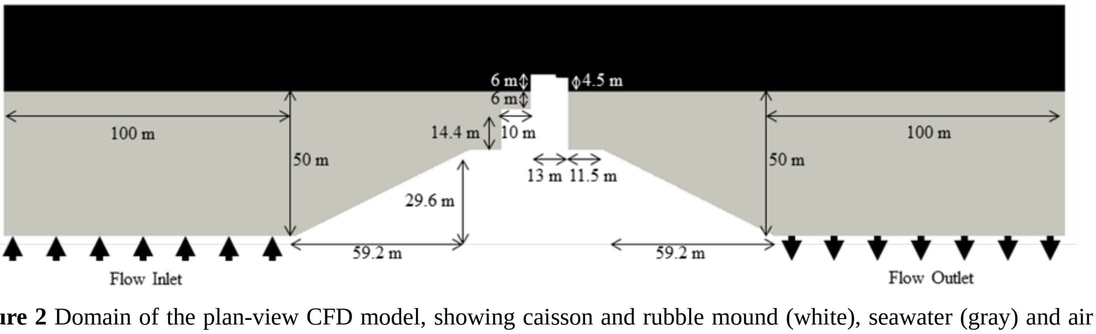

Figure 2 Domain of the plan-view CFD model, showing caisson and rubble mound (white), seawater (gray) and air (black). Seawater level shown is Mean Lower Water, which is the reference level for water levels discussed in this paper. Seaward is to the left, and landward is to the right. For force calculations, the length of the domain into the paper is assumed to be 30 m, which is the actual caisson length in that direction.

However, Arikawa et al's experiments did not consider the possibility of rubble mound geotechnical failure. Takagi (2009) showed that one of the most common modes of caisson breakwater damage during storms is punching failure of the rubble mound foundation (Goda, 2010). Furthermore, Yaoi et al (2012) ran centrifuge experiments and showed that foundation bearing capacity failure may have occurred during overtopping of deep-water caissons.

The work presented here uses a 2-dimensional shallow water equation model to reproduce the actual overtopping conditions experienced by the Kamaishi breakwater. The water levels seaward and landward of the breakwater are then used as boundary conditions in a 2-dimensional profile-view volume of fluid (VOF) computational fluid dynamics (CFD) model to analyze the forces experienced by the caisson and its rubble mound foundation. The result shows that punching failure of deep-water caissons was a likely mode of failure of these caissons. As such, scour prevention measures (such as armor blocks), though necessary to prevent scour-induced caisson failure, may not be sufficient to prevent geotechnical failure of deep-water caissons during overtopping.

2 METHODS

2.1 Shallow Water Modeling of Tsunami

Behavior of the tsunami near the Kamaishi breakwater was simulated using the Delft-3D hydrodynamic model, operating in two-dimensional depth-averaged mode. Domain decomposition was used, with the coarsest grid having 2 km resolution, an intermediate grid with 200 m resolution, and a fine grid with 20 m resolution (Figure 1). 500-m resolution bathymetry data were used in the open ocean (Japan Oceanographic Data Center, 2012), while 50-m resolution bathymetry data were used for Kamaishi Bay (Ministry of Land, Infrastructure, and Transport, 2012). Topography was 50-m mesh survey data (GSI, 2012).

Bottom roughness was assumed to be uniform everywhere, with a Manning's n of 0.02. Since the goal of this simulation was to replicate the behavior of the wave near the breakwater, and not the inundation experienced on land, the use of a uniform Manning's n was assumed to be sufficient. The 2km-resolution Pacific domain utilized the Reimann boundary condition in Delft-3D. In order to minimize reflection of the tsunami wave from the boundaries, the grid was oriented parallel to the tsunami wave front. The model time step was 0.3 sec, in order to ensure stability.

Initial conditions consisted of a water surface elevation near mean low water. In the 2-km resolution Pacific domain, the tsunami source used was based on that of Saito et al (2011). However, the source of Saito et al (2011) alone produced a tsunami near Kamaishi smaller than that measured by PARI (2012). As such, Saito et al's (2011) source was modified so as to be slightly larger (similar to the source determined by Takahashi et al, 2011), and to extend further to the northeast. The resulting source water elevation used is shown in Figure 1. Figure 3 shows a comparison of the tsunami water level resulting from this source, with the water level observed by the Port and Airport Research Institute (2012) at the three buoys closest to Kamaishi. The modeled water surface elevation is in good agreement with the measured water surface elevation.

The tsunami behavior was simulated for 2 hours. Since observations showed the breakwater caissons to have been displaced in the landward direction, it is likely they failed during the incident tsunami wave. Figure 4 shows the spatially average elevation of the simulated water surface on each side of the breakwater, starting at the time when overtopping began (the crest of the breakwater was 6 m above mean low water) until after the crest of the tsunami had passed. The maximum modeled water surface elevation on the seaward side of the breakwater was about 13 m, and the maximum water level difference across the breakwater was 9 m. This is in agreement with the numerical shallow water simulations discussed in Arikawa et al (2012).

2.2 CFD modeling of breakwater overtopping

In order to model forces on the caisson and its foundation, the OpenFOAM CFD model was used. Within OpenFOAM, the InterFOAM VOF module was applied, so as to simulate both air (density 1 kg/m^3) and water (density 1020 kg/m^3). Figure 2 shows the two-dimensional profile-view model domain. For purposes of calculating forces on the caisson, the domain was assumed to be 30 m long (normal the paper). Within the domain, the caisson was a no-slip hydraulically rough boundary with roughness length 1 mm, while the rubble mound was a region of porous flow with porosity 0.4, and Darcy-Forcheimer coefficients α_0=1500 and β_0=3.6 (per Mitsui et al, 2012). The left and right sides of the domain were free-slip walls. The top surface was open to the atmosphere. Turbulence was modeled using the standard Reynolds Averaged Navier Stokes k-ε formulation.

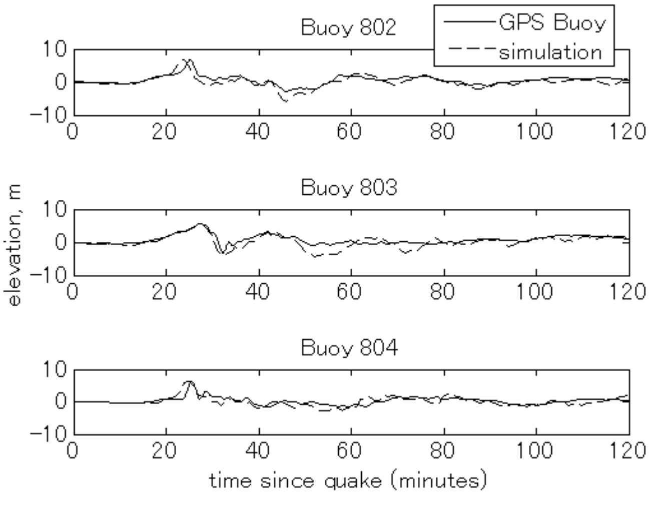

Figure 3 Comparison of measured and modeled (Delft-3D) water levels at the Port and Airport Research Institute's GPS buoy locations near Kamaishi.

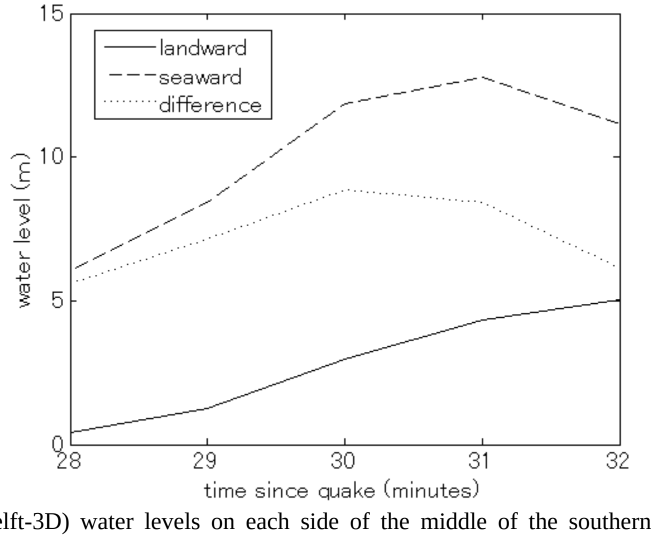

Figure 4 Modeled (Delft-3D) water levels on each side of the middle of the southern section of the Kamaishi breakwater, as well as the difference in water level across the breakwater, during the incident tsunami wave.

Simulating a laboratory flume, water was allowed to upwell and drain through the seabed on each side of the mound (as shown in Figure 2), with the flow rates set so as to reproduce the water levels of Figure 4 on each side of the caisson. Setting up the model in this way (instead of using water level and flow speed boundary conditions in place of the side walls) enhanced model stability.

The model consisted of a variable size rectangular grid, with grid cell area shown in Figure 5. In the region of the overtopping jet, grid cell dimensions were as small as 2 cm by 2 cm, while far away from the area of energetic flow grid cells size increased up to 1 m by 1 m. Landward and seaward of the rubble mound (the location of inflow and outflow), horizontal model resolution was coarsened to 10 m. The lower resolution on the sides of the model domain was implemented in order to reduce model run time. Since flow velocity was quite small in these regions, they were not of concern for the overtopping flow or calculation of force on the caisson. The model time step was varied automatically, in order to assure Courant number stability, but was typically on the order of 0.001 sec. Also, in order to preserve a well-defined free surface, anti-diffusion (interface compression) was applied to the free surface.

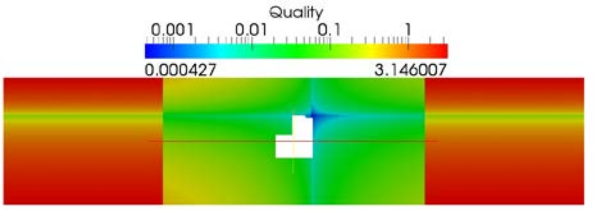

Figure 5 Area of model grid cells in m^2. Since the cells are rectangular, cell dimension is approximately the square root of the area. Grid is finest in region of overtopping flow. Note that the porous rubble mound is part of the grid, while the solid caisson is not.

3 RESULTS

Figure 6 shows the simulated caisson overtopping. Two important phenomena are visible here. First, the overtopping jet, even though gaining buoyancy due to entrained air, strikes the rubble mound with speeds up to 8 m/s, corroborating the conclusion of Arikawa et al (2012) that scour due to the overtopping jet was a cause of caisson failure. Second, the pressure behind the overtopping jet is significantly reduced from the hydrostatic pressure, due to the overtopping jet pulling further and further away from the landward side of the caisson as the overtopping water level increases, and due to the eddy that forms behind the jet.

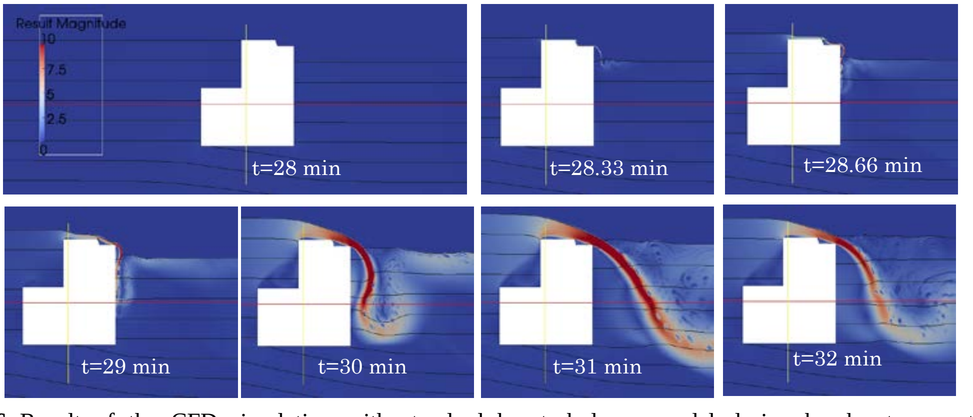

Figure 6 Result of the CFD simulation with standard k-ε turbulence model during breakwater overtopping. Pressure contours in the water are shown at 50 kPa intervals, with the free surface at a gauge pressure of 0 kPa. Color is water flow speed in m/s. Seaward is to the left, and landward is to the right.

3.1 Calculation of forces on caisson

Pressure is calculated by OpenFOAM at each grid cell along all surfaces of the caisson. This includes the lower surface, as beneath the caisson, the rubble mound is modeled as a porous body (per Mitsui et al, 2012), so flow through the mound, and the water pressure acting on the bottom surface of the caisson, is also resultant from the model. The net horizontal and vertical forces on the caisson due to water pressure are determined by equations (1) – (3). The overturning moment about the heel of the caisson due to water pressure is given by equation (4).

$$\vec{F}_k = P_k \, dA_k \, \hat{n}_k \tag{1}$$
$$F_{drag} = \sum \vec{F}_k \cdot \hat{\imath} \tag{2}$$
$$F_{lift} = \sum \vec{F}_k \cdot \hat{\jmath} \tag{3}$$
$$M_{overturning} = -\sum \vec{r}_k \times \vec{F}_k \tag{4}$$

in which F_k: force acting on the k^{th} cell along the caisson surface, P_k: pressure acting on the k^{th} cell along the caisson surface, dA_k: area of the k^{th} cell along the caisson surface, $\hat{n}_k$: unit vector normal to the k^{th}

cell along the caisson surface, F_{drag}: total drag force acting on the caisson, $\hat{i}$: horizontal unit vector, F_{lift}: total lift force acting on the caisson, $\hat{j}$: vertical unit vector, $M_{overturning}$: overturning moment acting about the heel of the caisson, and r_k: vector pointing from the heel of the caisson to the k^{th} cell along the caisson surface.

To find the punching pressure on the rubble mound foundation at the heel of the caisson (Goda, 2010), the normal force W_e on the caisson due to the rubble mound foundation is found by equation (5)

$$W_e = mg - F_{lift} \tag{5}$$

where m is the mass of the caisson, and g is the acceleration due to gravity (the effect of buoyancy is already included in the term F_{lift}). The moment M_e about the heel of the caisson due to the normal force is given by equation (6)

$$M_e = mgt - M_{overturning} \tag{6}$$

where t is the horizontal distance between the heel of the caisson and its center of mass. The horizontal distance t_e between the heel of the caisson and the location of the resultant normal force W_e is given by equation (7)

$$t_e = \frac{M_e}{W_e} \tag{7}$$

The punching pressure P_e at the heel of the caisson is then calculated by equation (8)

$$P_e = \begin{cases} \frac{2W_e}{3t_e} & : t_e \leq B/3 \\ \frac{2W_e}{B}\left(2 - 3\frac{t_e}{B}\right) & : t_e > B/3 \end{cases} \tag{8}$$

where B is the width of the caisson (23 m as shown in Figure 2).

3.2 Resulting forces on caisson

Figure 7 shows the calculated drag force on the caisson as a function of time during overtopping. The solid line shows the result of the CFD model. The dotted line shows the hydrostatic drag force, which is the horizontal force the caisson experiences due only to the difference in water level across the caisson. The difference between these two lines is the non-hydrostatic force on the caisson, mostly due to the suction induced on the landward side of the caisson as the overtopping flowrate increases and the overtopping jet pulls further and further away from the caisson (this suction is evident as dips in the pressure contours on the landward side of the caisson in Figure 6). The dashed line represents the frictional force resisting caisson sliding, which was calculated assuming a coefficient of friction between the caisson and the rubble mound of 0.6 (Goda, 2012) and a uniform bulk density of 1900 kg/m3 for the caisson and its fill. Figure 7 shows that the horizontal force on the caisson was never strong enough to cause sliding.

Figure 8 shows the lift force on the caisson. As before, the hydrostatic component (dotted line) is due to the buoyancy of the caisson itself. The full lift force (solid line) is slightly greater than the caisson buoyancy due to dynamic lift above the caisson during overtopping, but this effect is minimal. The lift force is much smaller than the weight of the caisson (dashed line).

Figure 9 shows the overturning moment about the (landward) heel of the caisson. As before, the hydrostatic component of this moment (dashed line) is due to the horizontal force on the caisson resulting from the difference in water level across it. The full overturning moment (solid line) is larger than the hydrostatic moment because of the suction discussed above. Nonetheless, toppling of the caisson due to the overturning moment is not possible, because the restoring moment due to the caisson’s weight (dashed line) is larger than the overturning moment.

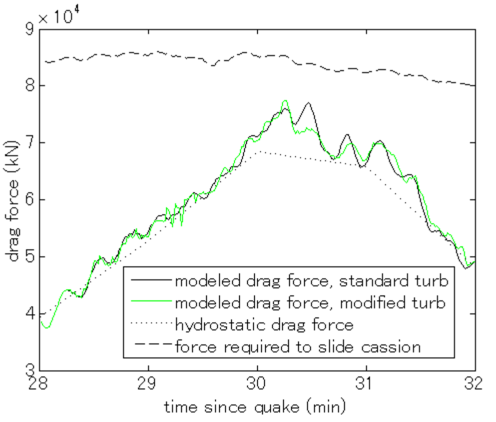

Figure 7 Horizontal force on caisson, for a caisson length normal to the model domain of 30 m.

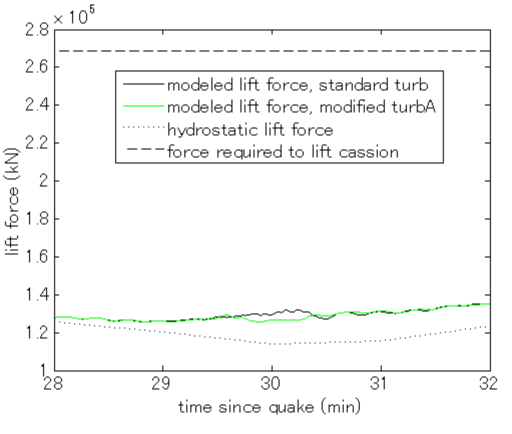

Figure 8 Lift force on caisson, for a caisson length normal to the model domain of 30 m.

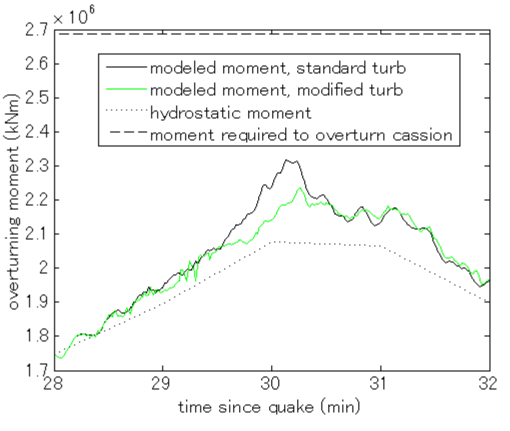

Figure 9 Overturning moment on caisson, for a caisson length normal to the model domain of 30 m.

Figure 10 shows the punching pressure at the heel of the caisson. The hydrostatic component (dotted line), due to the water level difference across the caisson, exerts a punching pressure of up to 900 kPa, while the total punching pressure (solid line) reaches up to 1200 kPa. The allowable design punching pressure (dashed line) is 600 kPa (Goda, 2010), while Uezono and Odani (1987) showed that a critical punching pressure of 800 kPa caused rubble mound foundation failure during experiments on similar structures in the field. Figure 10 shows that the Kamaishi breakwater caissons experienced a punching pressure much greater than either the design or critical value, for about 2 minutes.

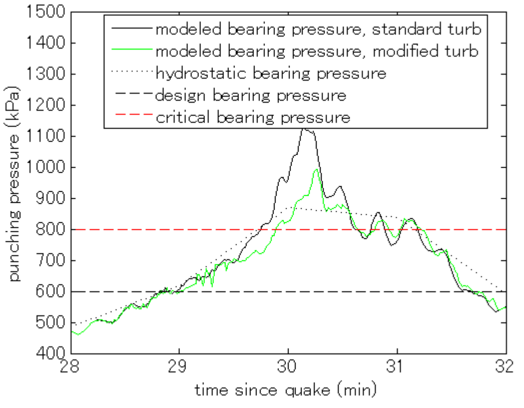

Figure 10 Bearing stress at the heel of the caisson, for a caisson length normal to the model domain of 30 m.

4 COMPARISON WITH EXPERIMENTS

In order to verify the CFD model results, qualitative comparison of overtopping jet's behavior with the experiments of Mitsui et al (2012) was carried out. Mitsui found that the InterFOAM VOF simulation overestimates the rate at which air trapped under the overtopping jet is entrained into the jet, thus causing the jet to spuriously pull close to the landward wall of the breakwater. This is evident in Figure 6 at t=28.66 minutes and t=29 minutes, where the jet has already pulled right up against the caisson and points vertically downward. In the simulation the air that had existed below the overtopping jet was rapidly entrained into the jet, and since there is no path for new air to replace the entrained air, the jet had to pull up against the caisson wall. In experiments, however, much of the air under the overtopping jet remains in place, and so the jet never pulls right up against the caisson.

The reason for the spurious entrainment of air in the simulation is due to the turbulence model implemented. In InterFOAM, the standard k-ε formulation is implemented, but lacking the buoyant production term. In this model, the kinematic eddy viscosity is continuous across the free surface, resulting in a large diffusive flux of momentum between water and air. However, Nakayama & Yokojima (2003) showed that in reality the kinematic eddy viscosity in water rapidly decreases to nearly zero at the air-water interface, because turbulent eddies from the water do not jump up into the air, and eddies in the air do not dig down into the water surface. Thus, simulations in which the eddy viscosity does not decay to zero at the air-water interface experience too much flux of momentum and scalars across the interface.

To account for the decay of turbulence at the air-water interface, another simulation was run with the k-ε turbulence model modified such that the turbulent eddy viscosity was set to zero in the air, defined as cells with $\alpha<0.1$, where α is the phase fraction of water ($\alpha=0$ indicates a cell with only air, while $\alpha=1$ indicates a cell with only water). In effect, this reduced the flux of momentum across locations with an air-water interface.

Figure 11 shows the effect of this modified turbulence model on the overtopping jet. At times up to and including t=29 min, air still exists under the overtopping jet, so the jet has not pulled all the way up against the landward wall of the caisson. This is close to the behavior seen in the experimental results of Mitsui et al (2012), an improvement over the result of the standard turbulence model with continuous kinematic eddy viscosity shown in Figure 6.

Figure 11 shows that the modified turbulence model results in reduced suction as the overtopping jet pulls away from the caisson as compared to the standard turbulence model, but that the maximum bearing force is still greater than that of the hydrostatic case. The maximum bearing stress of 1000 kPa with the modified turbulence model is less than the 1200 kPa resulting from the standard turbulence model, but is still large enough to cause foundation failure according to Uezono and Odani (1987).

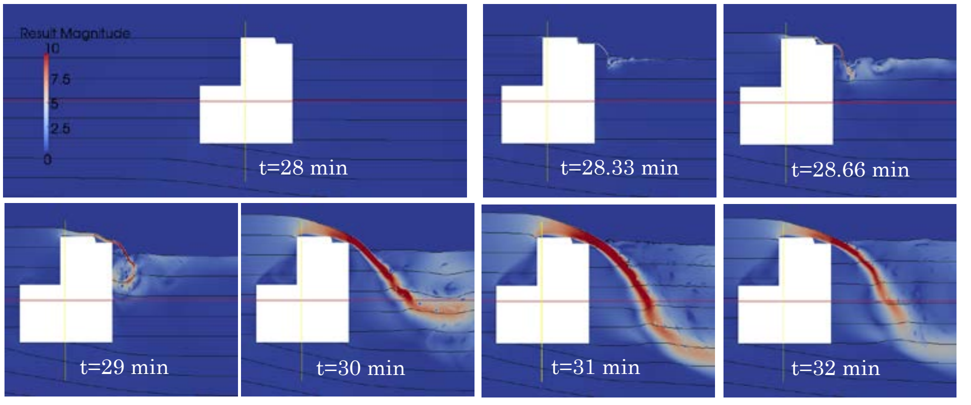

Figure 11 Result of the CFD simulation with modified k-ε turbulence model (zero eddy viscosity in air) during breakwater overtopping. Pressure contours in the water are shown at 50 kPa intervals, with the free surface at a gauge pressure of 0 kPa. Color is water flow speed in m/s. Seaward is to the left, and landward is to the right.

5 CONCLUSIONS

Numerical simulations were used to evaluate potential failure modes of the Kamaishi breakwater during the 2011 tsunami. In addition to the scour reported by Arikawa et al (2012), punching failure of the rubble mound foundation is shown to be a major cause of breakwater failure. The amount of punching stress hindcast by the CFD simulation is greater than the critical value of 800 kPa reported by Uezono and Odani (1987). Using a standard turbulence model with continuous kinematic eddy viscosity across the air-water interface, the maximum hindcast punching stress is 1200 kPa, while a turbulence model with zero eddy viscosity in the air, which better reproduces overtopping behavior as observed in laboratory experiments, hindcasts a maximum punching stress of 1000 kPa. Even though the modified turbulence model results in smaller bearing stress than the standard model, it is still larger than the critical bearing stress for foundation punching failure.

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References

Arikawa, T., Sato, M., Shimosako, K., Tomita, T., Tatsumi, D., Yeom, G., and Takahashi, K. (2012). “Investigation of the failure mechanism of Kamaishi breakwaters due to tsunami: initial report focusing on hydraulic characteristics.” Technical note No. 1251 of the Port and Airport Research Institute. Yokosuka, Japan (in Japanese).

Goda, Y. (2010). “Random seas and design of maritime structures”, 3rd edition. World Scientific. New Jersey. Chapter 4.

GSI (Geospatial Information Authority of Japan, 2012). 50 m mesh digital elevation map for eastern Japan. CD-ROM.

Japan Oceanographic Data Center (accessed January 2012). http://jdoss1.jodc.go.jp/cgi-bin/1997/depth500_file.jp

Ministry of Land, Infrastructure, and Transport (data received May 2012). Sendai Research and Engineering Office.

Mitsui, J., Maruyama, S., Matsumoto, A., and Hanzawa, M. (2012). "Stability of armor units covering harbor-side rubble mound of composite breakwater against tsunami overflow." Journal of Japan Society of Civil Engineers, Ser. B2 (Coastal Engineering), Vol.68, No.2, pp.I_881-I_885 (in Japanese).

Nakayama, A., and Yokojima, S. (2003). “Modeling free-surface fluctuation effects for calculation of turbulent open-channel flows”. Environmental Fluid Mechanics. 3: 1-21.

Port and Airport Research Institute (accessed January 2012). GPS buoy data. http://nowphas.mlit.go.jp/nowphasdata/sub300.htm

Saito, T., Ito, Y., Inazu, D., and Hino, R. (2011). “Tsunami source of the 2011 Tohoku-oki earthquake, Japan: inversion analysis based on dispersive tsunami simulations”. Geophysical Research Letters, vol. 38, L00G19.

Takagi, H. (2009). “Estimation of stability performance of existing breakwaters using a reliability approach and proposal of effective countermeasures to prevent future failures”. Proceedings of the Japan Society of Civil Engineers B, Vol. 65, No. 3, 246-258 (in Japanese).

Takahashi, S., Toda, K., Kikuchi, Y., Sugano, T., Kuriyama, Y., Yamazaki, H., Nagao, T., Shimosako, K., Negi, T., Sugeno, H., Tomita, T., Kawai, H., Nakagawa, Y., Nozu, A., Okamoto, O., Suzuki, K., Morikawa, Y., Arikawa, T., Iwanami, M., Mizutani, T., Kohama, E., Yamaji, T., Kumagai, K., Tatsumi, D., Washizaki, M., Izumiyama, T., Seki, K., Yeom, G., Takenobu, M., Kashima, H., Banno, M., Fukunaga, Y., Sakunaka, J., and Watanabe, Y. (2011). “Urgent survey for 2011 Great East Japan Earthquake and Tsunami disaster in ports and coasts.” Technical Note No. 1231 of the Port and Airport Research Institute. Yokosuka, Japan (in Japanese).

Uezono, A., and Odani, H. (1987). “Planning and construction of the rubble mound for a deep water breakwater: the case of the Kamaishi bay mouth breakwater.” Chapter 4.1 of Coastal and Ocean Geotechnical Engineering. The Japanese Geotechnical Society. Tokyo, Japan (in Japanese).

Yaoi, Y., Iai, S., and Tobita, T. (2012). “Centrifuge tests and analysis of complex breakwater failure due to tsunami.” Proceedings of the Japan Society of Civil Engineers Annual Congress. 435-436. Nagoya, Japan (in Japanese).